CONTACT-AWARE NUCLEUS SEGMENTATION

Varun Kumar Dasoju[1], *Qingsu Cheng*[2], *Zeyun Yu*[1]

University of Wisconsin–Milwaukee, Milwaukee, WI, USA
[1]Department of Computer Science; [2]Department of Biomedical Engineering
vdasoju@uwm.edu, chengq@uwm.edu, yuz@uwm.edu

ABSTRACT

Pixel overlap can conceal the biologically important merging of touching nuclei. We extend an EfficientNet-B7/UNet++ pipeline with an auxiliary head that predicts only interfaces between adjacent nuclei, while a foreground head preserves complete masks. Markers obtained by removing predicted contacts are expanded inside the foreground to recover instances without eroding outer contours. On the official BBBC039 split, ten-seed mean Dice was unchanged (0.9660 versus 0.9657), while aggregated Jaccard index (AJI), panoptic quality (PQ), and object F1 increased from 0.6833/0.7434/0.7545 to 0.8703/0.8610/0.8865. The corresponding paired image differences were significant after Holm correction. On the fixed NucleusSegData split, nucleus/pixel F1 reached 0.9315/0.8791 for Huh7 and 0.8809/0.8189 for HepG2, exceeding directly comparable published point estimates. Equal-budget controls and uncertainty across ten seeds support contact prediction, rather than edge preprocessing, as the main contribution.

Index Terms— nucleus instance segmentation, fluorescence microscopy, contact boundaries, Gabor filters, reproducibility

1. INTRODUCTION

Nuclear masks support cell counting, morphology, and spatial analysis, but dense touching nuclei are difficult to annotate and separate. U-Net [1], UNet++ [2], and pretrained encoders are strong semantic baselines, while biological instance methods emphasize boundaries or object geometry [3, 4, 5]. A high foreground Dice score is insufficient when one predicted component merges several cells.

We study a mammary-cell segmentation pipeline combining multi-scale Gabor responses, EfficientNet-B7, UNet++, and spatial/channel squeeze-and-excitation (SCSE). A controlled audit found that the Gabor bundle improves semantic Dice only slightly over vanilla B7 and does not exceed a matched Xception/SCSE model. Metric-matched evaluation then exposed the larger bottleneck: connected components preserve foreground overlap but merge touching nuclei, reducing AJI, PQ, and object F1.

Our main contribution is a contact-aware formulation that (1) preserves a full-foreground objective, (2) learns only interfaces between touching instances rather than all outer contours, and (3) reconstructs complete instances by constrained marker expansion. We evaluate it on the provider-defined BBBC039 split using ten seeds, native-grid semantic and instance metrics, paired uncertainty, equal-budget controls, and public code. We then use the fixed Huh7/HepG2 NucleusSegData split as an independent public-domain check. A private MCF10A set is retained only as supplementary internal-domain evidence.

2. METHOD

2.1. Input representation and backbone

Let x be percentile-rescaled grayscale intensity. The optional edge channel is the maximum absolute response of 24 L1-normalized Gabor filters,

$$e(x) = \mathcal{N}\left(\max_{\theta,f} \left| x * \frac{k_{\theta,f}}{\|k_{\theta,f}\|_1 + \epsilon} \right| \right), \tag{1}$$

using eight orientations, $f \in \{0.05, 0.15, 0.25\}$ cycles/pixel, kernel size 31, $\sigma = 5$, aspect ratio 0.5, and zero phase. A learned $4 \to 16 \to 8 \to 3$ projection permits ImageNet initialization. EfficientNet-B7/UNet++/SCSE predicts either one semantic logit or two contact-aware logits. Vanilla B7 and UNet++/Xception/SCSE are matched controls.

2.2. Contact-aware instance formulation

BBBC039 annotations encode touching-aware instances. For neighboring pixels u, v with instance labels $g(u), g(v)$, the contact target is

$$c(u) = \mathbb{1}\{g(u) > 0,\ g(v) > 0,\ g(u) \neq g(v)\}, \tag{2}$$

aggregated over the eight-neighborhood and dilated by one pixel inside the foreground. Unlike a complete contour target, c excludes nucleus–background edges. The network predicts foreground p_f and contact probability p_c and minimizes

$$\begin{aligned} \mathcal{L} = \tfrac{1}{2}[\mathcal{L}_D(p_f, y_f) + \mathcal{L}_B(p_f, y_f)] \\ + \tfrac{1}{4}[\mathcal{L}_D(p_c, c) + \mathcal{L}_{B,w}(p_c, c)]. \end{aligned} \tag{3}$$

where $\mathcal{L}_D$ is soft Dice, $\mathcal{L}_B$ is binary cross-entropy, and $\mathcal{L}_{B,w}$ compensates contact sparsity with a finite batchwise positive weight. At inference, connected components of $\hat{y}_f \setminus \hat{c}$ form markers. Every foreground pixel is assigned to its nearest marker; thus contacts separate instances without removing foreground area. Post-processing parameters are selected without access to test labels.

3. EXPERIMENTAL DESIGN

3.1. Data and splits

BBBC039v1 contains 200 16-bit 520×696 Hoechst images of U2OS cells and approximately 23,000 expert-annotated nuclei [6, 7]. We use the official 100/50/50 train/validation/test partition. No crops or ROIs are generated. Image identifiers, decoded-image hashes, and split manifests are checked for overlap. Validation contains one empty reference; test contains none. Empty/empty pairs score overlap one and one-empty pairs score zero.

NucleusSegData contains 2,661 manually annotated nuclei in 37 Hoechst images from Huh7 and HepG2 liver-cancer cell lines [8, 9]. We convert its RGB-encoded files deterministically to grayscale and retain the published split: 785 nuclei in ten training images, 891 nuclei in 11 Huh7 test images, and 985 nuclei in 16 HepG2 test images. The test subsets remain untouched during model and threshold selection. This benchmark is independent of BBBC039 and closely matches the grayscale fluorescence image formation assumed by our model.

BBBC039 was manually delineated by expert biologists, and NucleusSegData by the providers' biologist collaborator. Neither release provides repeat delineations suitable for inter-annotator agreement; consequently, ground truth is treated as a single reference rather than uncertainty-free anatomy. This limitation is kept separate from seed variability.

The private UWM lineage contains 855 annotated 16-bit MCF10A confocal planes (Zeiss LSM, 40×/1.1 W, 405/461 nm, 0.2506 μm pixels, 1.0 μm z-step). One trained annotator delineated foreground with expert review; repeat-annotation agreement is unavailable. Only 334 planes retain links to eight source stacks, while 521 lack acquisition identifiers. We therefore use 23 conservative geometry-defined groups split into 516/133/206 planes (11/5/7 groups) and report this set as internal evidence, not independent validation. All 855 masks contain foreground. Private images are neither distributed nor needed to reproduce either public result.

3.2. Training and endpoints

On BBBC039, all matched models use 512^2 inputs, batch size 8, 20 epochs, AdamW (3×10^{-4}, weight decay 10^{-4}), cosine decay, gradient clipping at one, flips, and 90° rotations. Ten seeds quantify optimization variability. The highest validation object F1 selects contact-aware checkpoints; no test image is used for tuning. Predictions are restored to the native grid. On NucleusSegData, the same formulation is fitted to all ten designated training images for 60 fixed epochs (batch size 2); the final epoch is always used. For each seed, contact threshold $\{.1, \ldots, .7\}$ and minimum component size $\{16, 32, 64, 128\}$ are selected by nucleus F1 on the training images only and then locked for Huh7 and HepG2 inference. The UWM primary and vanilla controls use ten matched seeds, 30 epochs, batch size 8, and validation-Dice checkpoint selection; inference uses no test-time augmentation.

Semantic endpoints are foreground Dice (DICE1), IoU, boundary F1, surface Dice at two pixels, and HD95. Instance endpoints reproduce published definitions: ensemble DICE2, AJI, detection/segmentation/panoptic quality (DQ/SQ/PQ), symmetric best Dice, and Caicedo object F1 averaged over IoU thresholds 0.50–0.90 [3]. We average seeds within each test image, use 10,000 paired bootstrap resamples for 95% confidence intervals, and apply two-sided paired Wilcoxon tests with Holm correction. Seed SD is reported as optimization variability, not biological sample size.

Table 1. Official BBBC039 test results, mean (seed SD), $n = 10$. Literature rows use the same split and published metric definitions.

Model/output	DICE1	AJI	PQ	Obj.-F1
Vanilla semantic + CC	.9657(.0014)	.6833(.0096)	.7434(.0059)	.7545(.0074)
Gabor semantic + CC	.9659(.0035)	.6776(.0116)	.7419(.0069)	.7525(.0067)
Vanilla contact-aware	**.9660(.0023)**	**.8703(.0064)**	**.8610(.0049)**	**.8865(.0044)**
Gabor contact-aware	.9652(.0045)	.8614(.0122)	.8558(.0111)	.8788(.0142)
ASW-Net + expansion [4]	.9645	.9020	.8665	–
BISSG [5]	.9482	.8680	.8629	–
Caicedo U-Net [3]	–	–	–	.8980

3.3. Reproducibility safeguards

Vanilla and Gabor contact models have 68.16 and 68.35 million parameters, respectively, and receive identical update budgets. Test-time augmentation is disabled. The sparse-contact positive weight is clipped to $[1, 50]$ with an explicit zero-positive safeguard, and non-finite losses stop training. Configuration, initial-state, checkpoint, archive, and decoded-image hashes are recorded; a completed run is rejected if its protocol hash differs. Per-image predictions and metrics are retained so aggregate results can be independently reconstructed. These checks are covered by automated tests and are released with the code.

4. RESULTS

Table 1 confirms that overlap alone hides instance merging. Relative to matched vanilla semantic connected components, contact prediction changed Dice by only 0.0003 (95% CI $[-.0002, .0007]$, Holm $p = .333$), but increased AJI by 0.1870 ($[.1597, .2156]$), PQ by 0.1176 ($[.0998, .1366]$), and

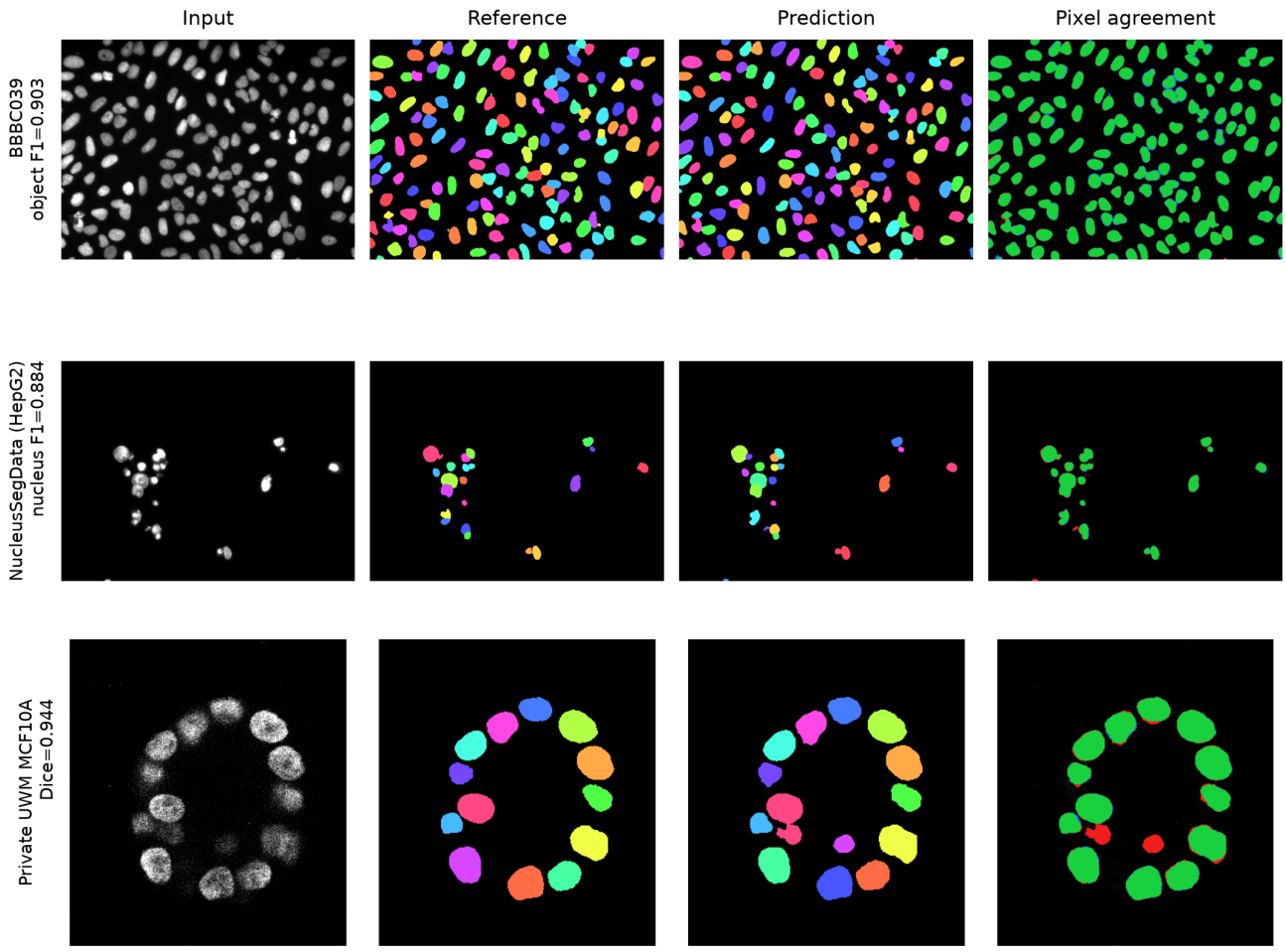


Fig. 1. Real held-out outputs across the complete evidence chain. Public examples use seed 4, the median-performing seed, and images nearest that seed's median performance; the private example is the median held-out image for seed 0 of the ten-seed Gabor/SCSE semantic study. Instance colors are identifiers, not classes. Agreement maps show true-positive (green), false-positive (red), and false-negative (blue) pixels. Displayed scores are image-specific; aggregate results are in Table 2. The UWM labels are semantic and cannot train the contact target.

object F1 by 0.1320 ($[.1119, .1535]$); all three instance gains had Holm $p < 3 \times 10^{-14}$. Detection quality rose from .8187 to .9402, whereas segmentation quality changed from .9060 to .9141, localizing the gain primarily to separation. Vanilla contact-aware B7 also exceeded its Gabor counterpart on AJI by .0089 ($p_{\text{Holm}} < 5 \times 10^{-8}$), so the simpler model is preferred. Cross-paper significance is not claimed because repeated literature predictions are unavailable.

The completed semantic audit found ten-seed Dice of 96.57% for vanilla B7, 96.59% for the Gabor/projection/SCSE bundle, and 96.86% for Xception/SCSE. Filter-bank, Sobel, neutral-edge, boundary-loss, residual-fusion, and weighted-sampling controls changed Dice by less than 0.2 points around the reference in the developmental screen. A failed full-contour three-class control is also retained: even after boundary expansion, seed 0 obtained .8889 DICE1, .6749 AJI, .6962 PQ, and .6098 object F1. This result motivated predicting contacts only. Developmental selections are not used for confirmatory significance; the final table uses complete ten-seed runs.

Table 2 separates comparable endpoints rather than pooling unlike metrics. On NucleusSegData, our nucleus/pixel F1 exceeds the strongest published row by 2.40/3.93 points on Huh7 and 3.88/5.44 points on HepG2. On BBBC039, our DICE1 is 0.15 points higher than ASW-Net, but AJI, PQ, and object F1 remain 3.17, 0.55, and 1.15 points below the best published point estimates; thus we claim a substantial matched-baseline gain, not overall state of the art. On private UWM data, ten-seed Gabor/SCSE Dice is 0.35 points below the matched vanilla control (paired seven-group difference $-.0035$, 95% bootstrap CI $[-.0064, -.0004]$; Wilcoxon $p = .0781$), providing no evidence of superiority at $\alpha = .05$.

Table 2. Unified benchmark interpretation. Results are mean (seed SD), $n = 10$; "prior" is the strongest directly comparable published point estimate identified for each public endpoint. Δ is ours minus prior/control in percentage points. UWM is private, so its comparator is a matched internal control. Cross-paper significance is not claimed.

Dataset	Endpoint	Ours	Protocol/uncertainty	Comparator	Prior	Δ pp
BBBC039	DICE1	.9660(.0023)	10 seeds; 50 images	ASW-Net+expansion [4]	.9645	+0.15
	AJI	.8703(.0064)	same	ASW-Net+expansion [4]	.9020	−3.17
	PQ	.8610(.0049)	same	ASW-Net+expansion [4]	.8665	−0.55
	Object F1	.8865(.0044)	same	Caicedo U-Net [3]	.8980	−1.15
NucleusSegData Huh7	Nucleus F1	**.9315(.0284)**	10 seeds; 11 images	Object-oriented [9]	.9075	**+2.40**
	Pixel F1	**.8791(.0364)**	same	Object-oriented [9]	.8398	**+3.93**
NucleusSegData HepG2	Nucleus F1	**.8809(.0347)**	10 seeds; 16 images	Object-oriented [9]	.8421	**+3.88**
	Pixel F1	**.8189(.0413)**	same	Object-oriented [9]	.7645	**+5.44**
Private UWM MCF10A	Dice	.9319(.0033)	10 seeds; 7-group CI [.9144,.9484]	Vanilla B7/UNet++ (internal)	.9354(.0013)	−0.35

Published per-image predictions and repeated-run uncertainty are unavailable, so literature comparisons remain descriptive.

5. DISCUSSION AND CONCLUSION

The results support the central hypothesis that explicit contact prediction converts high semantic overlap into better instance separation. Predicting only nucleus–nucleus contacts avoids the foreground shrinkage observed with full-contour targets, while constrained expansion restores complete shapes. Ten seeds, paired image inference, and replication on a second fixed-split benchmark show that the gain is not confined to one favorable initialization.

The study also limits the Gabor claim: its small semantic advantage is matched by a lower-parameter architecture, and contact-aware vanilla B7 is stronger and more stable than contact-aware Gabor B7 over ten seeds. NucleusSegData tests a second acquisition and two cell lines, but both public benchmarks remain fluorescence cell-line data. Its ten-image training split also produces appreciable seed variability, which we expose rather than treating test images as tuning data. Further limitations are incomplete source provenance for the private lineage and no repeat-annotation agreement study. Future work should add multi-laboratory images and compare fine-tuned generalist models under known pretraining exposure.

The evidence is therefore tiered rather than pooled: BBBC039 is the confirmatory benchmark with validation-based model selection, NucleusSegData is fixed-split external-domain replication, and UWM is internal sensitivity evidence with conservative grouping. This separation prevents the larger private collection from compensating for uncertain source provenance and prevents unmatched literature summaries from entering statistical tests.

6. COMPLIANCE WITH ETHICAL STANDARDS

BBBC039 and NucleusSegData are openly released cell-line benchmarks. The supplementary private data contain an established MCF10A cell line and no human participants or identifiable information; institutional review board approval and informed consent are therefore not applicable.

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